

GRASSY LAKE, AB, Canada

WMO: 717880

Lat: 49.8736N

Lon: 111.7328W

Elev: 783

StdP: 92.27

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.8	-24.6	-31.1	0.2	-27.6	-27.6	0.3	-24.4	14.3	0.4	13.1	-3.3	3.7	140	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	32.3	17.5	30.3	17.0	28.4	16.4	19.4	28.2	18.3	27.2	17.5	26.1	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.8	23.0	15.2	11.9	21.7	14.2	11.1	20.5	58.6	28.0	55.1	27.0	52.1	26.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.4	10.0	DB	-32.4	35.6	3.4	1.6	-34.9	36.8	-36.9	37.7	-38.8	38.6	-41.3	39.8	(4)
(5)			WB	-32.6	21.9	3.4	1.3	-35.0	22.8	-37.0	23.5	-38.9	24.2	-41.3	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-6.0	-7.1	0.2	5.8	11.6	16.0	19.5	18.4	13.6	6.7	-0.4	-7.0	(6)	
	DBStd	11.28	8.36	8.79	7.58	5.22	4.23	3.31	2.98	3.28	4.77	5.22	7.19	8.00	(7)	
	HDD10.0	2444	497	480	307	140	32	2	0	0	19	126	314	528	(8)	
	HDD18.3	4628	756	713	564	375	212	86	22	40	153	361	562	786	(9)	
	CDD10.0	985	0	0	2	15	81	180	293	261	128	23	2	0	(10)	
	CDD18.3	128	0	0	0	0	3	15	57	43	11	0	0	0	(11)	
	CDH23.3	2252	0	0	0	12	106	267	887	711	259	11	0	0	(12)	
	CDH26.7	803	0	0	0	2	22	72	349	269	89	1	0	0	(13)	
(14) Wind	WSAvg	4.8	5.5	4.7	5.2	5.3	4.8	4.5	3.8	3.7	4.3	4.9	5.2	5.1	(14)	
(15) Precipitation	PrecAvg	364	17	13	19	30	47	83	36	34	31	21	15	14	(15)	
	PrecMax	532	44	29	50	68	102	207	125	80	68	70	32	32	(16)	
	PrecMin	193	6	2	1	7	7	15	3	2	0	6	2	0	(17)	
	PrecStd	76	10	7	12	15	24	41	28	21	20	14	9	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.3	18.3	25.0	29.3	32.1	34.9	33.7	32.6	24.9	18.2	10.7	(19)	
		MCWB	5.5	6.2	8.8	11.5	14.1	18.7	18.8	17.6	16.1	13.3	9.8	4.6	(20)	
	2%	DB	8.3	8.5	15.9	21.1	26.4	28.6	32.7	31.8	29.5	20.8	13.9	7.6	(21)	
		MCWB	3.8	3.7	7.5	10.4	13.2	16.3	18.6	17.0	15.2	11.2	7.1	3.5	(22)	
	5%	DB	6.3	6.1	13.2	18.0	23.6	26.3	30.4	29.9	26.4	17.8	11.3	5.5	(23)	
		MCWB	2.7	2.5	6.1	8.4	12.1	15.6	18.1	16.6	14.2	9.7	6.1	2.3	(24)	
	10%	DB	4.5	4.2	10.4	15.3	20.9	24.2	28.4	27.6	23.2	15.2	8.9	3.8	(25)	
		MCWB	1.8	1.3	4.6	7.1	11.0	14.6	17.3	16.1	13.0	8.4	4.3	1.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	6.7	9.6	12.2	14.8	19.5	21.5	19.8	17.4	14.0	10.0	5.6	(27)	
		MCDB	10.5	12.2	17.3	22.2	26.5	27.3	30.5	29.0	27.5	21.8	17.5	9.2	(28)	
	2%	WB	4.1	4.1	8.2	10.5	13.7	17.9	20.0	18.4	16.0	11.8	7.8	3.7	(29)	
		MCDB	7.9	7.9	14.9	20.3	23.8	26.6	28.8	28.0	26.7	19.4	13.1	6.9	(30)	
	5%	WB	2.9	2.7	6.5	8.9	12.8	16.6	19.0	17.6	14.8	10.2	6.3	2.4	(31)	
		MCDB	5.8	5.9	12.7	16.7	21.7	24.5	27.8	27.1	24.7	16.8	10.9	5.3	(32)	
	10%	WB	1.8	1.4	4.8	7.4	11.8	15.4	18.1	16.7	13.5	8.9	4.7	1.1	(33)	
		MCDB	4.4	4.0	9.9	14.4	19.6	22.7	26.5	25.7	22.6	14.8	8.5	3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.9	12.0	13.1	13.5	13.4	15.6	15.7	14.7	12.9	10.9	9.9	(35)	
		MCDBR	10.5	12.1	15.6	18.5	18.4	17.0	19.4	19.9	20.2	16.6	13.9	10.6	(36)	
	5% WB	MCWBR	7.1	8.5	8.7	8.9	7.7	7.2	7.9	7.7	8.2	8.5	8.2	7.4	(37)	
		MCWBR	10.0	11.7	15.3	17.2	16.0	15.9	17.1	17.5	19.2	15.6	13.5	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.265	0.271	0.295	0.344	0.363	0.354	0.355	0.377	0.334	0.292	0.265	0.255	(40)	
		taud	2.435	2.475	2.464	2.321	2.316	2.394	2.406	2.314	2.415	2.514	2.494	2.421	(41)	
	Ebn at Noon		802	892	923	900	894	903	896	855	863	852	800	761	(42)	
		Edh at Noon	60	74	90	116	122	114	111	116	94	71	56	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.25	2.24	3.56	4.92	5.83	6.40	6.98	5.60	3.99	2.48	1.39	0.97	(44)	
	RadStd		0.11	0.18	0.26	0.34	0.43	0.39	0.35	0.35	0.33	0.25	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page